

The Use of Indirect Positive Evidence in Second Language Morphological Learning by Different First Language Learners

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Abstract

This study investigates the impact of learners' first language on their detection of indirect positive evidence presented in their second language while learning the plural morphology of an additional foreign language. The goal is to explore an alternative approach to learning foreign language grammars. The participants were 43 native Thai-speaking and 42 native Spanish-speaking learners of Mandarin who had advanced proficiency in second language English. They were divided into experimental and control groups and learned 12 Mandarin nouns in their singular form over 2 consecutive days. Only the experimental group was exposed to indirect positive evidence that provided evidence of the unmarked plural feature in Mandarin. After the learning phase, participants' knowledge of the learned words was tested using singular and plural prompts. Follow-up questions were asked to investigate the basis of participants' responses. The results showed that six Thai and three Spanish participants detected and used the indirect positive evidence. This finding suggests that language learners differ in their ability to adopt and use indirect positive evidence as a function of their first language background.

Keywords

indirect positive evidence, adult learners, second/foreign language learning

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1. Introduction

Although the direct input of grammatical rules has been widely adopted in language learning (R. Ellis, 1999), indirect positive evidence (IPE) has shown potential effectiveness in foreign language learning. IPE refers to evidence drawn from errors that native speakers of a second language (L2) make in the learners' first language (L1) (Schwartz & Goad, 2017). For example, Li and Goad (2024) exposed native English speakers with no prior knowledge of Mandarin to Mandarin speakers' incorrect production of L2-English plurals, in which the plural inflection -s was intentionally omitted. This provided indirect evidence of the lack of plural inflection in Mandarin nouns. It was found that approximately one-third of the English speakers successfully detected the target feature. Nevertheless, it remains unclear whether IPE can be detected and effectively used when presented in learners' L2 rather than their L1, and whether IPE can be better detected and used by learners whose L1 contains the target grammar than by those whose L1 lacks it. To fill this knowledge gap, the present study compared L1-Thai and L1-Spanish advanced L2-English speakers' learning of Mandarin plurals via IPE—specifically, Mandarin speakers' production of incorrect English plural nouns that omitted the plural inflection -s. Thai and Spanish speakers were selected because plural inflection exists in Spanish but not in Thai. It was anticipated that the findings of this study would shed further light on IPE usage in the learning of a foreign language's morphology. From a pedagogical perspective, adult language learners frequently struggle with foreign morphology associated with functional categories (e.g., Goad & White, 2004; Goad et al., 2003). The findings are expected to have pedagogical implications for the optimal use of IPE in adult foreign language learning and teaching.

2. Literature Review

In conventional foreign language classrooms, grammar is typically taught by directly imparting the rules and features of the target language to students (R. Ellis, 1999). However, adult learners often face challenges in achieving native-like performance, particularly in morphology associated with functional categories (e.g., Jiang, 2004, 2007; Lardiere, 1998a, 1998b). Alternative learning approaches are therefore essential.

Goad and colleagues demonstrated the efficacy of IPE in foreign language learning, initially in L2 phonology. In Schwartz and Goad (2017), native English speakers were exposed to IPE of phonotactic constraints on word-initial clusters in Brazilian Portuguese. In a dialogue between two Brazilian Portuguese speakers in Brazilian Portuguese-accented English, sC clusters were produced as epenthesized [i]+sC. Participants were then asked to produce novel English words with the same accent. Eleven out of 32 (34%) participants correctly judged sC-initial words ill-formed in Brazilian Portuguese and repaired the clusters with protheses following the dialogue pattern, demonstrating their ability to use the available IPE.

Building on this, Li and Goad (2024) investigated whether naïve English-speaking learners of Mandarin could detect the omission of English plural inflections by Mandarin speakers and infer that plurality is not overtly marked in Mandarin. Specifically, native English speakers with no prior knowledge of Mandarin were trained to learn Mandarin

countable nouns with singular referents (pictures showing the corresponding meaning of a word with a single object) on 2 consecutive days. At the beginning of each day, only the exposure group was exposed to a 4-minute dialogue between a native Mandarin speaker and a native English speaker, in which the Mandarin speaker incorrectly produced 24 nouns in obligatory plural contexts as singulars. At the end of the study, both groups were tested for word-learning knowledge using singular referents, whereas their use of IPE was tested using plural referents (pictures showing the corresponding meaning of a word with multiple objects). Three follow-up questions were asked to investigate how decisions were made in response to multiple referents. The results showed that both groups learned 12 words well (singular accuracy: 87.5%–100%). In response to plural referents, 12 participants (exposure group: $n = 10$; control group: $n = 2$) showed accuracy of 75%–95.8%, and the rest showed accuracy ranging from 0% to 25%. Nevertheless, their responses to the follow-up questions indicated that eight participants from the exposure group discovered and used IPE, while the remaining four based their decisions on analogy with singulars. Overall, 40% of participants in the exposure group successfully detected the target IPE (lack of plural inflections in Mandarin) and used it in the test.

Despite the success of these studies, both Li and Goad (2024) and Schwartz and Goad (2017) focused on native English speakers exposed to IPE in their L1, English. Real-world scenarios may involve IPE presented in learners' L2 rather than their L1. Consider a Japanese student and a Spanish student, both unfamiliar with French, studying non-French courses taught in English in Paris. The exposure to French-accented English in their daily interactions with classmates, teachers, and/or friends may lead to their discovery of certain French grammatical features, with potential variation based on L1 differences. Specifically, this raises the question of whether Japanese learners can detect French grammatical features based on errors made by native French speakers in their English production.

Hypothesizing that morphological similarities between learners' L1 and a foreign language facilitate the learning of the morphology of the target foreign language, whereas morphological differences between two languages hinder the learning process as revealed by some scholars (e.g., Al-Qadi, 1992; Azaz, 2019; Ramírez et al., 2010, 2011), we might expect Spanish students to outperform Japanese students because Spanish shares more inflectional features with French than does Japanese.

This hypothesis aligns with the widely accepted notion that morphological similarities between learners' L1 and the target foreign language facilitate learning, whereas differences may impede the learning process (Ramírez et al., 2010, 2011). For instance, Arabic speakers may face challenges in English derivations due to significant morphological differences (Al-Qadi, 1992). Similar observations extend to third language (L3) studies, where language similarities positively impact grammatical learning (Cenoz, 2001; Gass & Selinker, 1992; Kellerman & Sharwood Smith, 1986; Odlin, 1989; Schepens et al., 2016). For example, Schepens et al. (2016) conducted a lexical and morphological test among 39,300 L3-Dutch speakers of diverse L1s and L2s. The results showed that larger L1–L3 and L2–L3 distances were correlated with lower degrees of L3 learnability.

Theoretical frameworks supporting this view include Jiang et al.'s (2011) Morphological Congruency Hypothesis (MCH), which defines two languages as

morphologically incongruent if a grammatical morpheme exists in one language but is absent in the other, and as morphologically congruent if a grammatical morpheme is present in both. Learning an L2 that is morphologically congruent with learners' L1 is easier than learning an L2 that is morphologically incongruent with their L1.

Goad and colleagues' Prosodic Transfer Hypothesis (PTH) analyzes language learners' failure to correctly produce L2 morphological inflections, such as word-final inflectional morphemes, from the perspective of prosody. According to PTH, prosodic constraints restrict the types of representations that can be built for L2 learning, consequently limiting L2 learners' production of the inflectional morphology associated with functional categories (Goad & White, 2004; Goad et al., 2003). Mandarin speakers' omission of inflectional morphemes in English (e.g., plural inflections, third-person singular inflections, and past tense inflections of verbs) can be explained, according to PTH, by the fact that Mandarin lacks coda clusters (Goad & White, 2008, 2009).

Similarly, Lado's (1957) Contrastive Analysis Hypothesis (CAH) systematically explores the influence of L1 on L2 learning and divides this influence into positive transfer, negative transfer, and zero transfer. It predicts that positive transfer can facilitate L2 learning when learners' habitual L1 responses are similar to new L2 features; negative transfer may hinder L2 learning when L1 habitual responses differ from those of the L2; and zero transfer poses difficulty when L1 habitual responses have no relationship with those of the L2 (see also R. Ellis, 1985). Based on MCH and CAH, prior studies revealed that Chinese speakers undersupplied English plurals and/or verbs in obligatory contexts (e.g., Bialystok & Miller, 1999; Lardiere, 1998a, 1998b). This may be attributed to a lack of overtly marked inflections in Chinese, which makes Chinese–English morphologically incongruent (see MCH) and/or to negative transfer effects (see CAH) caused by the difference.

Additionally, learners' prior language experiences, as proposed by Koda's Transfer Facilitation Model, indicate that metalinguistic awareness¹ from prior literacy resources can aid L2 learning (Koda, 2008, 2021). Nevertheless, due to cross-linguistic similarities and variations in morphological structures across languages, learners' morphological awareness in biliteracy acquisition is governed by the typological distance between their L1 and the target foreign language (Zhang, 2013). For example, Wang et al. (2006) examined Korean children's acquisition of English literacy and found that phonological awareness in participants' L1 Korean transferred and facilitated L2 English reading, but this was not the case for orthographic processing. This is explained by the fact that both Korean and English are alphabetic languages but differ in how phonemic information is represented in orthography: phonemes in English are presented orthographically in a linear string of letters, but they are combined in a nonlinear order in Korean (Wang et al., 2006). Likewise, Wang et al. (2009) found that Chinese–English bilingual children transferred Chinese phonological awareness to English word reading, whereas such transfer was not found in the processing of English orthography, which was attributed to the contrasting writing systems of the two languages. Overall, it is presumed that a morphological feature present both in learners' L1 and in the target foreign language is more transferable than features absent in their L1.

Given that the present study investigated L1-Thai and L1-Spanish speakers' learning of Mandarin via IPE provided in English, it raises questions about the potential influence of L2 English on participants' learning of Mandarin. Previous research suggests that

bilinguals/multilinguals benefit from rich language learning experiences; hence they can think about language in a more abstract way (Jessner, 2008; Moore & Coste, 2006; Ransdell et al., 2006), develop higher levels of cognitive ability (Bialystok, 2010), and enhance metalinguistic awareness (Cenoz, 2001; Jessner, 2012, 2014). For instance, benefiting from prior language experiences, adult learners initially acquire a foreign language on an explicit/declarative basis before they can process it automatically (N. C. Ellis, 2005; Falk et al., 2015; Paradis, 2009; Ringbom, 2007). This is in line with the Cumulative Enhancement Model (CEM) (Flynn et al., 2004), which indicates that the effect of a previously learned L2 is either neutral or positive for learning an additional foreign language.

Take Li and Goad (2024) as another example: most of their participants (18 out of 20 in the exposure group) had some knowledge (not above intermediate proficiency) of one or more languages in addition to their L1. To some extent, the high success rate (accuracy: 87.5%–100%) in the trained Mandarin words might be attributed to their rich language learning experience. Accordingly, the bilingual participants in the present study were also expected to learn the target Mandarin words.

Moreover, learners with rich L2 learning experience might have encountered difficulties such as grammatical mistakes caused by L1–L2 differences. As a result, when exposed to mistakes/errors made by other speakers (IPE) that were similar or identical to their own, they were expected to detect the mistakes/errors and infer that the speakers' L1 might have the same or similar grammatical rules as their L1. In comparison, those whose L1 is similar or identical to the target L2 in a given grammatical area may not have had such experience; hence, they may be less likely to detect this IPE.

These theoretical frameworks lay the foundation for the present study. Specifically, the linguistic feature affordance of IPE is determined by typological similarity or distance between learners' L1 and L2, which either supports or impedes morphological transfer. The pedagogical affordance of IPE concerns learners' likelihood of detecting such evidence and transferring the inferred feature(s) to support the acquisition of target L2 feature(s). By comparing L1-Thai and L1-Spanish advanced L2-English speakers, this study examines how these two affordances interact when IPE is presented in the learners' L2.

3. Plurality in English, Spanish, Mandarin, and Thai

This section compares plurality marking in the four languages central to the study: English, Spanish, Thai, and Mandarin. These languages were selected because they create a clear typological contrast in plural morphology and because plurality is a functional category that adult L2 learners frequently struggle with. Mandarin and Thai speakers commonly omit the English plural suffix *-s* in obligatory contexts and provide ideal indirect positive evidence for the absence of plural marking in Mandarin.

In English, the plural meanings of countable nouns are commonly realized by adding the inflectional morpheme *-s* to the right edge of the noun's singular form (e.g. *cake* → *cakes*). The suffix *-s* can be replaced with *-es* (or *-ies*) when nouns end in sibilants or the sounds /s, z, ʃ, tʃ, ʒ, or dʒ/ (e.g. *bus* → *buses*; *country* → *countries*) (Davenport & Hannahs, 2010; Teschner, 2007). These are usually referred to as regular English plurals. Exceptions related to English plural marking are beyond the scope of this study. Similarly,

most countable nouns in Spanish form plurals by adding *-s* or *-es* to the word-final position of the noun's singular form. Specifically, *-s* is added to nouns ending in unstressed vowels and stressed *-é* (e.g. *chico* → *chicos*; *café* → *cafés*), while *-es* is added to those ending in stressed vowels (except *-é*) and consonants (e.g. *pared* → *paredes*; *rey* → *reyes*) (Saporta, 1965). Nevertheless, the plural inflections *-s/-es* in English are phonetically realized as /s/ or /z/ depending on the phonetic context, but they are only produced as /s/ in Spanish.

In comparison, plural inflection does not occur in Mandarin or Thai. Both Mandarin and Thai words are internally unchangeable; hence, bare nouns can be interpreted as singular or plural, and plural marking is not obligatory whenever determiner phrases bear plural denotations in the two languages. Specifically, plural functions in Mandarin are realized through context rather than inflections by: (a) using a numeral before a noun (e.g., *xué shēng* (student/students) → *qī gè xué shēng* (seven students)); (b) adding a determiner before a noun (e.g., *nà xiē xué shēng* (those students)); or (c) adding the collective marker *-men* after a noun (e.g., *xué shēng men* (students)), where *-men* is a lexical item with its own semantics and distribution rather than an inflectional suffix like *-s* or *-es* in English and Spanish (Teschner, 2007). In Thai, singular nouns are pluralized by: (a) adding a classifier and/or numerical designator after a noun (e.g., *thoorasàp* (telephone/telephones) → *thoorasàp (telephone) sām* (three) *khŕaay* (classifier)); (b) placing the noun *phúak* (group, party) before a noun (e.g., *dèk* (child/children) → *phúak-dèk* (the children)); or (c) reduplication, which is lexically restricted to human nouns (e.g., *dèk* (child/children) → *dèk-dèk* (the children)) (Jenks, 2011).

In summary, English and Spanish require overt suffixation for plural nouns, whereas neither Mandarin nor Thai uses inflectional marking. This contrast—L1-Spanish being congruent with English plurals and L1-Thai being congruent with Mandarin—creates differential linguistic and pedagogical affordances for detecting IPE presented in L2 English. These cross-linguistic patterns directly ground the research questions and predictions in the following section.

4. Research Questions and Predictions

The central inquiry driving this study is whether IPE provided in learners' L2 during L3 grammar learning can be more effectively detected and utilized by learners whose L1 contains the target grammar feature than by those whose L1 does not. This study focuses particularly on the domain of inflectional morphology. To address this, two research questions (RQs) are proposed:

RQ1: Can L1-Thai and L1-Spanish participants detect the IPE of Mandarin plural features provided in L2-English material?

Prediction: there could be three possible outcomes:

1. There is no significant difference in the ability of L1-Thai and L1-Spanish participants to detect the IPE of Mandarin plural features in L2-English material.

2. L1-Thai participants, whose L1 contains the target grammar feature, will demonstrate a significantly higher ability to detect the IPE compared to L1-Spanish participants, whose L1 lacks the target grammar feature.
3. L1-Spanish participants outperform L1-Thai participants.

Drawing from insights in Li and Goad (2024), where 15 out of 20 participants (75%) reported noticing multiple errors in exposure material presented in their L1 (English), the present study diverges because all participants are advanced L2 English speakers. It is hypothesized that their ability to identify plural errors in the same exposure material will not surpass that of native English speakers. Consequently, their detection rate is expected to be lower than that observed in Li and Goad (2024).

RQ2: Can those who detected the IPE on Mandarin plural features form the exposure material use it in the test?

Prediction: three outcomes might occur:

1. There is no significant difference between L1-Thai and L1-Spanish groups in the ability of participants who detected the IPE of Mandarin plural features to utilize it in the tes
2. Participants from the L1-Thai group, with potential prior experience in learning regular English plurals, will demonstrate a significantly higher ability to utilize the detected IPE in the test compared to participants from the L1-Spanish group.
3. Participants from the L1-Spanish group outperform the L1-Thai group.

Considering that neither Thai nor Mandarin plurals exhibit overt inflectional marking, Thai and Mandarin English learners often underproduce regular English plurals in obligatory contexts (Hiranburana, 2017; Jiang et al., 2011). Thai participants' experiences or errors in learning regular English plurals may lead them to infer that Mandarin learners of English make similar errors. This inference could facilitate the detection of the target IPE. In contrast, regular Spanish plurals share similarities with regular English plurals in terms of inflectional changes (e.g., the addition of *-s* or *-es* to the right edge of a singular form). Moreover, languages that realize grammatical meanings through inflections have been reported to better facilitate learning English functional inflections than languages whose words are internally unchangeable (e.g., Hwang & Lardiere, 2013; Jiang et al., 2011). Accordingly, Spanish speakers were assumed to be less likely to have experienced the same or similar difficulties in learning regular English plurals compared to Thai speakers. Therefore, it was predicted that Spanish speakers might be less likely to detect the target IPE than Thai participants.

5. Methodology

This study involved a training process adapted from Li and Goad (2024) and a post-training test. The experiment was programmed using PsychoPy v3.0 (Peirce et al., 2019).

All recordings were made in a soundproof booth with a Roland R-05 recorder (settings: 16-bit mono channel, 44.1 kHz) and saved in WAV format. The study employed a mixed-methods design: quantitative measures assessed participants' performance in the training and testing phases, while qualitative follow-up questions examined whether and to what extent the experimental groups detected the IPE provided in the exposure material.

5.1. Participants

The participants were native L1-Thai and L1-Spanish learners of L2 English. As described in Section 3, plural formation is similar between Thai and Mandarin (no overt inflections) and between Spanish and English (regular plurals formed by adding word-final inflections). Specifically, 43 Thai and 42 Spanish learners of English were recruited based on the following criteria: (a) advanced L2 English proficiency International English Language Testing System (IELTS) score of 7.5 or above; (b) age between 18 and 30 years with an age of onset of L2 English learning between 5 and 8 years; and (c) no knowledge of any language other than their L1 and English. All participants were university students majoring in non-linguistics fields at neighboring universities in England. They reported no knowledge of Chinese² and no hearing, speaking, or other language-related difficulties. All participants volunteered for the study, signed an ethical consent form prior to participation, and received compensation. They were free to withdraw at any time (no one dropped out).

One concern might arise here: can L2 English speakers successfully detect IPE provided in L2 English rather than in their L1? To minimize this potential bias, all participants in the study were advanced L2 English speakers. A pilot study³ conducted prior to the main study found that advanced L2 English speakers were able to notice some of the errors provided as IPE. Moreover, as reported in previous studies on inflectional morphology, L2 English learners with advanced proficiency displayed native-like sensitivity to English plural errors in self-paced reading tasks despite the lack of the inflectional plural feature in their L1 (Li, 2021).

The participants were randomly divided into experimental and control groups (with gender balance maximized across groups): Thai experimental (Thai-ex; $n = 22$, 13 females, 9 males), Spanish experimental (Spanish-ex; $n = 21$, 12 females, 9 males), Thai control (Thai-con; $n = 21$, 13 females, 8 males), and Spanish control (Spanish-con; $n = 21$, 11 females, 10 males). Analysis of variance (ANOVA) revealed no significant differences among the groups in age, gender, age of onset, or years of L2 English study ($p > .05$).

5.2. The Experiment

Following Li and Goad (2024) (see Table 1), the experiment lasted 2 consecutive days for each participant. The experimental groups listened to the exposure material (a dialogue) once at the beginning of each day. They then learned the target words in three mini-sessions, each followed by a mini-test on word-learning knowledge. Maxi-test⁴ 1 was administered at the end of Day 1 to assess the 12 learned words. Day 2 followed the same procedure, except that maxi-test 1 was replaced with maxi-test 2, which examined participants' detection and use of the IPE provided in the exposure material. The control

Table 1. Schedule of the Experiment (adapted from Li & Goad, 2024).

	Experimental groups	Control groups
Day 1	Step 1: Listen to dialogue	Step 1: Mini-learning 1
	Step 2: Mini-learning 1	Step 2: Mini-test 1
	Step 3: Mini-test 1	Step 3: Mini-learning 2
	Step 4: Mini-learning 2	Step 4: Mini-test 2
	Step 5: Mini-test 2	Step 5: Mini-learning 3
	Step 6: Mini-learning 3	Step 6: Mini-test 3
	Step 7: Mini-test 3	Step 7: Maxi-test 1
	Step 8: Maxi-test 1	
Day 2	Repeat Steps 1–7 from Day 1	Repeat Steps 1–6 from Day 1
	Step 8: Maxi-test 2	Step 7: Maxi-test 2
	Step 9: Follow-up questions 1–4	Step 8: Follow-up questions 3–4

groups followed the same procedure but did not listen to the exposure material. Follow-up questions (a brief survey) were administered at the end of the study.

5.2.1. Exposure Material. The exposure material (see Appendix) was a 718-word interlocutor-balanced dialogue (Mandarin speaker: 354 words; English speaker: 364 words; 19 turns each), adapted from Li and Goad (2024). The stimulus words providing IPE were the same 24 nouns produced in obligatory plural contexts, all of which were intentionally uttered in singular forms by the native Mandarin speaker. The dialogue was recorded by a phonetically trained native English speaker (a 33-year-old male with a PhD in linguistics) and a native Mandarin speaker (a 23-year-old female working in finance in England, IELTS 6.0). To demonstrate that the Mandarin speaker’s omission of word-final /s/ or /z/ in plurals was not due to an inability to produce these sounds, other non-plural s-shaped clitics at the right edge of monomorphemic words were produced error-free. The recording was delivered at a natural speed and lasted 4:02 minutes.

5.2.2. Learning Sessions. In the learning sessions, participants were introduced to the same set of 12 pseudo-Mandarin words used by Li and Goad (2024). These monosyllabic Mandarin words followed either a consonant-vowel-vowel (CVV) or consonant-vowel-nasal (CVN) structure and were produced with Tone 4. Tone 4 was chosen to minimize tonal influences on participants’ performance, given its familiarity to Thai speakers and its relatively natural realization for non-tonal language speakers such as Spanish participants (Li & Goad, 2024).

To match the words with their meanings, pseudo-referents were used because of the difficulty in finding authentic monosyllabic Mandarin words in Tone 4 that corresponded to countable nouns. Notably, none of the trained words, whether in singular or plural form, appeared in the Mandarin speaker’s utterances in the exposure material. Participants were not informed that the words were pseudowords until the experiment was completed.

Throughout the learning process, meanings were conveyed using pictures representing the singular form of the respective items (e.g. [pèi] (cup) shown in Figure 1). The 12 words were systematically divided into three 4-word mini-learning sessions. In each session, participants heard the auditory pronunciations of the target words paired with corresponding pictures, with each pair presented 3 times in random order, yielding 12 trials per mini-session. A mini-test then assessed participants' understanding of the just-learned four words. The test presented the auditory recording of a target word together with both the correct meaning (the same singular picture used during the learning sessions) and an incorrect meaning (a picture showing one of the other three words' singular referents) (see Figure 2).

Upon completing mini-test 3, a maxi-test on the 12 learned words followed, using the same methodology as the mini-tests. Each word was presented 3 times, yielding 36 trials in total. Participants' responses and the correctness of each trial were automatically recorded using PsychoPy (Peirce et al., 2019).

Day 2 replicated the procedure from Day 1, with maxi-test 1 replaced by maxi-test 2. In this test, two pictures were provided for each target word: one showing a single object (singular) and the other showing multiple objects (plural). Each trial presented the auditory form of a word with three on-screen options: (a) the correct meaning in either singular or plural form; (b) a randomly selected picture from another learned word (singular or plural form); and (c) "NEITHER," indicating that neither picture matched the heard word (e.g., Figure 3). There were 96 trials in maxi-test 2: 12 words $\times$ 2 presentations (singular and plural) $\times$ 4 different matches. Of these, 24 trials had plural-form pictures as the correct answer, 24 had singular-form pictures, and the remaining 48 had "NEITHER"⁵ as the correct answer.

5.2.3. Follow-Up Questions. The Thai-ex and Spanish-ex groups answered questions 1–4, whereas the Thai-con and Spanish-con groups answered only questions 3–4 because they did not listen to the exposure material:

1. Did you notice anything wrong with the female Mandarin speaker's utterances in the dialogue? If yes, what did you notice?
2. Did you notice any connection between the dialogue and maxi-test 2? If yes, what was the connection?
3. In maxi-test 2, when a learned word occurred in terms of a picture showing multiple objects related to its corresponding meaning, did you select it as a correct answer? Why or why not?
4. While learning English as an L2, did you have difficulty correctly producing regular English plurals (in writing or oral form)? If yes, (a) which type(s) of errors did you make? (b) Did you connect the plural errors you made with those in the exposure material?

6. Results

The primary focus of this study was on maxi-test 2 and the subsequent follow-up questions, which aimed to assess participants' ability to detect and utilize IPE provided in the



Figure 1. Sample picture used in the learning sessions for the target word [pèi] “cup.”



Figure 2. Sample picture used in mini-test sessions for target word [pèi] “cup.”



Figure 3. Sample picture used in maxi-test 2 for the target word [gòu] “book.”

exposure material. Maxi-test 1 served to obscure the purpose of the study and to further strengthen participants’ learning of the stimulus words in singular form. Individual participants’ accuracy rates were calculated as (number of correct responses ÷ number of testing stimuli) × 100%.

6.1. Results from Maxi-Test 2

Participants' accuracy rates on singular prompts (number of correct responses $\div$ 24 $\times$ 100%) were analyzed to confirm their mastery of the 12 target Mandarin words, a prerequisite for examining IPE use. Following Li and Goad (2024), participants who made more than three errors (accuracy below 87.5%) were excluded. Two males from the Thai-ex group, one male from the Spanish-ex group, and two females from the Spanish-con group were excluded, leaving 20, 20, 21, and 19 participants in the Thai-ex, Spanish-ex, Thai-con, and Spanish-con groups, respectively. As shown in Table 2 and Figure 4, mean singular accuracy exceeded 95% in all groups, with 51 of the 80 remaining participants achieving 100% accuracy (Thai-ex: $n = 11$; Spanish-ex: $n = 12$; Thai-con: $n = 13$; Spanish-con: $n = 15$). A one-way ANOVA with group as the between-subjects factor and singular accuracy as the dependent variable revealed no significant differences among the four groups ($F(3, 79) = 0.87$, $p = .46$, $\eta^2 = .34$).

Plural accuracy rates (number of correct responses $\div$ 24 $\times$ 100%) were examined to determine whether participants had detected and used the IPE provided in the exposure material when responding to plural prompts in maxi-test 2. Accuracy varied considerably across groups. As shown in Table 2 and Figure 4, both experimental groups showed a wide range of accuracies, with a subset of participants achieving high rates. Specifically, six participants in each experimental group achieved plural accuracy above 90%, with four Spanish-ex and two Thai-ex participants reaching 100% accuracy. However, the remaining participants in the two experimental groups showed 0% accuracy (Thai-ex: $n = 9$; Spanish-ex: $n = 8$), less than 10% (Thai-ex: $n = 5$; Spanish-ex: $n = 5$), or 12.5% (Thai-ex: $n = 0$; Spanish-ex: $n = 1$). In contrast, control group participants generally performed poorly: most showed 0% accuracy (Thai-con: $n = 14$; Spanish-con: $n = 13$), below 10% (Thai-con: $n = 7$; Spanish-con: $n = 4$), or 12.5% (Thai-con: $n = 0$; Spanish-con: $n = 1$), and none reached 100%. Nevertheless, two Spanish-con participants achieved accuracy above 90%, while no Thai-con participants did so. A univariate ANOVA was conducted with the individual participants' plural accuracy as the dependent factor, while *group difference* (experimental group vs. control group) and *L1* (Spanish vs. Thai) were coded as fixed factors. It was found that *group difference* played a significant effect on plural accuracy ($F(1, 79) = 9.73$, $p = .003$), whereas neither *L1* ($F(1, 79) = 0.53$, $p = .47$) nor its interaction with *group difference* ($F(1, 79) = 0.24$, $p = .63$) had a significant effect on plural accuracy. This suggests a link between exposure to IPE and performance in maxi-test 2.

NEITHER accuracy (number of correct responses $\div$ 48 $\times$ 100%), which reflected participants' responses to distractor prompts, showed consistent performance across groups and further corroborated their proficiency on singular prompts. As shown in Table 2 and Figure 4, the NEITHER accuracy for all groups ranged from 85% to 100%, with means above 95%. A one-way ANOVA revealed no significant differences among the four groups ($F(3, 79) = 0.97$, $p = .41$, $\eta^2 = .12$). This result aligns with their performance on singular prompts and confirms that the remaining 80 participants had a good command of the 12 learned words.

Table 2. Participants' Accuracy Rates (%) on Singular, Plural, and NEITHER Prompts in Maxi-Test 2.

Target answer	Group	Mean (%)	SE	Min. (%)	Max. (%)
Singulars	Thai-ex	97.5	0.7	91.7	100.0
	Spanish-ex	97.5	0.9	87.5	100.0
	Thai-con	97.4	0.8	87.5	100.0
	Spanish-con	98.9	0.5	91.7	100.0
Plurals	Thai-ex	29.8	9.9	0.0	100.0
	Spanish-ex	31.7	10.6	0.0	100.0
	Thai-con	1.8	0.6	0.0	8.3
	Spanish-con	11.2	6.7	0.0	95.8
NEITHER	Thai-ex	95.6	1.0	85.4	100.0
	Spanish-ex	97.8	0.7	89.6	100.0
	Thai-con	96.5	0.8	89.6	100.0
	Spanish-con	96.6	1.2	85.4	100.0

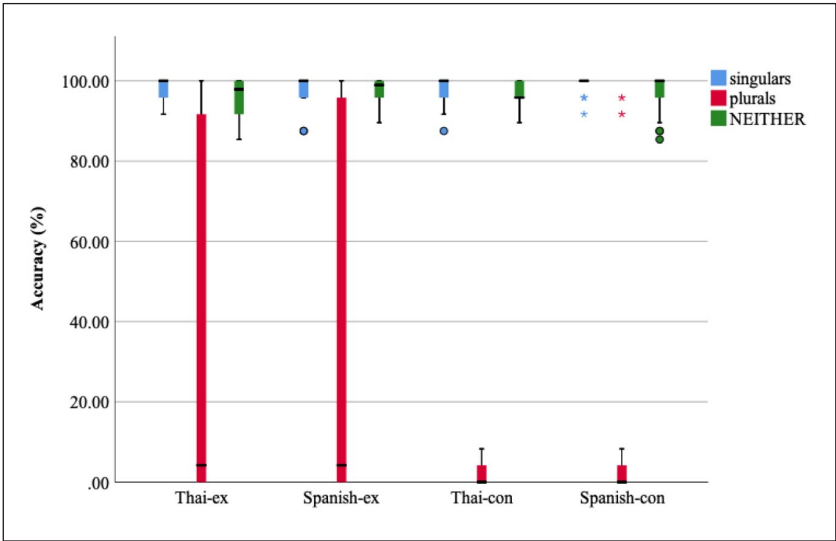


Figure 4. Boxplots of participants' performance (accuracy rates) in maxi-test 2.

6.2. Responses to Follow-Up Questions

1. Did you notice anything wrong with the female Mandarin speaker's utterances in the dialogue? If yes, what did you notice?

This question addressed participants' awareness of the female Mandarin speaker's errors in the exposure material. The majority of participants, including 15 (75%) Thai-ex and 12 (60%) Spanish-ex participants, reported noticing (some of) of the plural errors. Some

responses, however, were unrelated to the study's focus (e.g., problems with sentence tones or incorrect pronunciation of certain vowels/consonants).

2. Did you notice any connection between the dialogue and maxi-test 2? If yes, what was the connection?

Among the participants who reported noticing the plural errors in the exposure material, only six Thai-ex and three Spanish-ex participants discerned a connection with maxi-test 2. Remarkably, these nine participants exhibited plural accuracy rates ranging from 91.7% to 100%, suggesting that they had recognized the target IPE in the exposure material.

3. In maxi-test 2, when a learned word occurred in terms of a picture showing multiple objects related to its corresponding meaning, did you select it as a correct answer? Why or why not?

Participants' responses showed varied perceptions and strategies in maxi-test 2.

Thai-ex group: six participants affirmed selecting pictures showing multiple objects as correct answers, linking this to the idea that Mandarin nouns use the same form for singular and plural (those who answered "yes" to follow-up questions 1 and 2), as in Thai. The remaining 14 Thai-ex participants answered "no," stating that the plural pictures differed from those in the learning sessions or that they had learned only singular forms. Nevertheless, five of them selected the plural-form picture correctly in one or two trials (accuracy of 4.2% or 8.3%), attributing their responses to random guessing.

Spanish-ex group: six participants answered "yes." They also answered "yes" to the above two questions. However, only three reported inferring that Mandarin nouns use the same form for singular and plural meanings. The other three based their responses on analogy with the singular forms learned during training. Of the 14 participants who answered "no," six achieved plural accuracy of 4.2%–12.5% through guessing.

Thai-con group: 19 participants answered "no," and two of them said that they could not recall it clearly. Seven participants achieved plural accuracy of 4.2%–8.3%, citing responses through guessing or "could not remember" how they chose their response.

Spanish-con group: two participants answered "yes" (plural accuracy 91.7% and 95.8%), attributing their decisions to analogy. The remaining 17 answered "no," with four achieving 4.2%–8.3%. They reported that they "probably" responded with random guesses or could not respond.

Overall, six Thai-ex and three Spanish-ex participants successfully detected and applied the IPE in maxi-test 2. An additional three Spanish-ex and two Spanish-con participants relied on analogical reasoning. The remaining participants either consistently selected incorrect options for plural prompts or achieved only 1–3 correct trials through random guessing.

4. While learning English as an L2, did you have difficulty correctly producing (in writing or oral form) regular English plurals? If yes, (a) which type(s) of errors did you make? (b) Did you connect the plural errors you made with those in the exposure material?

Thai-ex and Thai-con groups: all participants reported making similar mistakes (omission of English plural inflections) or experiencing difficulties producing regular English plurals, particularly in spontaneous oral production.

Spanish-ex group: four participants reported making mistakes when speaking and/or writing regular English plurals, especially when their English proficiency was lower. Three of them indicated that they had detected and used the IPE. The remaining 16 participants reported that they had neither experienced such difficulties nor made mistakes similar to those of the Mandarin speaker in the exposure material.

Spanish-con group: three participants reported occasionally using singular forms in plural contexts when speaking English. The other 16 participants either answered “no” or indicated that they could not remember.

Table 3 provides a comprehensive summary of the four participant groups’ performance, shedding light on how the study’s RQs are addressed.

RQ1 focused on participants’ awareness of English plural errors produced by the Mandarin speaker in the exposure material. The results showed that 15 Thai-ex participants (75%) and 12 Spanish-ex participants (60%) demonstrated awareness of some of the English plural errors. This finding highlights the cross-linguistic influence on plural in L2 acquisition and participants’ attentiveness to relevant linguistic features.

RQ2 examined participants’ ability to detect and utilize IPE from the exposure material. The results showed that six Thai participants (30%) and three Spanish participants (15%) successfully detected and applied the IPE. This suggests that they might have developed a nuanced understanding of Mandarin pluralization patterns through inference from the IPE.

7. Discussion

The primary finding is that six Thai-ex (30%) and three Spanish-ex (15%) participants successfully detected and used the target IPE. Although descriptively more Thai than Spanish participants did so, statistical analysis showed no significant main effect of L1. This numerical difference should therefore be interpreted cautiously. Nevertheless, the overall results are consistent with the role of L2-English learning experience: all Thai participants reported difficulties producing English plurals (omission of *-s/-es*), whereas only a minority of Spanish participants did. This aligns with typological differences between Thai–English and Spanish–English plural morphology (Jiang et al., 2011).

Moreover, this finding aligns with the hypothesis that learners encounter challenges in L2 learning due to differences between their L1 and the target foreign language (Jiang et al., 2011; Lado, 1957). The results are also consistent with studies showing that Thai speakers often omit plural inflections in obligatory plural contexts in oral and/or written English (Hiranburana, 2017; Pongpaiboj, 2008; Sridhanyarat, 2013). Given that participants had no knowledge of any language other than their L1 and English and that none were linguistics majors, their L2 English learning experience was their only source of foreign language learning experience. Therefore, Thai participants’ relatively higher success rate in detecting the IPE of Mandarin plurals may, to some extent, be attributed to their L2 English learning experience. This is substantiated by the Thai-ex group’s responses to follow-up question 4, where all participants admitted making mistakes

Table 3. Summary of Four Groups of Participants’ Performance.

Groups	Range of plural accuracies	No. of participants who noticed plural errors in dialogue	No. of participants connected errors in dialogue to maxi test 2	Correct plural responses due to what?
Thai-ex (n=20)	0%–100%	16	6	IPE: <i>n</i> = 6 analogy: <i>n</i> = 0 guesses/could not remember clearly: <i>n</i> = 5
Spanish-ex (n=20)	0%–100%	18	3	IPE: <i>n</i> = 3 analogy: <i>n</i> = 3 guesses/could not remember clearly: <i>n</i> = 6
Thai-con (n=21)	0%–8.3%	N/A	N/A	IPE: N/A analogy: <i>n</i> = 0 guesses/could not remember clearly: <i>n</i> = 7
Spanish-con (n=19)	0%–95.8%	N/A	N/A	IPE: N/A analogy: <i>n</i> = 2 guesses/could not remember clearly: <i>n</i> = 4

IPE = indirect positive evidence.

when producing regular English plurals (using singular forms), with some still struggling with this error in spontaneous English production.

In contrast, only six Spanish-ex and three Spanish-con participants reported having or possibly having had difficulties producing regular English plurals (using singular forms in obligatory contexts). Interestingly, among the six Spanish-ex participants who reported such difficulties, half successfully detected the IPE and used it in maxi-test 2. Conversely, none of the Spanish-ex participants who claimed no difficulties (or could not remember having had any) managed to detect the target IPE. This finding, together with the Thai participants’ performance, supports the idea that L2 English learning experience may have facilitated detection of the target IPE provided in English.

Moreover, considering that both Mandarin and Thai lack plural inflections, the smaller number of Spanish participants who reported difficulties producing regular English plurals (compared with Thai participants) aligns with the hypothesis that cross-linguistic similarities facilitate the learning of a target foreign language, whereas differences may hinder it (e.g. Jiang et al., 2011; Lado, 1957; Ramírez et al., 2010). Exceptions, however, were observed in nine Spanish participants who indicated that they had difficulty correctly producing regular English plural inflections. Given that there were no significant

differences among participants in language background, it remains unclear what led to these distinctive experiences. Despite the uncertainty surrounding these negative L2 English learning experiences, it is possible that they contributed to the successful detection of IPE. Notably, the three Spanish-ex participants who successfully detected and used the IPE in maxi-test 2 all reported difficulties producing regular English plurals and achieved 100% accuracy on the plural prompts.

Building on this finding, a bold speculation can be made: although the grammar of Mandarin plurals was never directly taught, the six Thai-ex and three Spanish-ex participants successfully deduced its feature. This suggests that the grammatical element may not be particularly challenging for them if introduced directly. If true, the comparatively higher rate of IPE use by Thai participants (relative to Spanish participants) might indirectly support the notion that similarities between learners' L1 and a foreign language facilitate language learning (e.g. Jiang et al., 2011; Lado, 1957; Ramírez et al., 2010, 2011). This speculation warrants further investigation in future studies.

Another noteworthy finding concerns the Spanish participants. In addition to the three Spanish-ex participants who successfully detected the IPE, three more Spanish-ex and two Spanish-con participants achieved plural accuracy above 90% by relying on analogy with the singular forms learned during the training sessions. Li and Goad (2024) similarly reported that two participants in their exposure group used analogy when responding to plural items. Although their plural accuracy was somewhat lower than that of the three Spanish-ex participants in the present study (75%–87.5% vs. 95.8%–100%), none of the Thai participants reported using analogy. The reason for this discrepancy between Thai and Spanish participants remains unclear. This might be attributed to individual differences in cognitive abilities, nonverbal intelligence, and short-term memory (Brooks & Kempe, 2013; Brooks et al., 2017; R. Ellis, 2004).

Furthermore, although nine participants successfully detected the IPE in the present study, the rates (30% in the Thai-ex group and 15% in the Spanish-ex group) were lower than the 40% reported by Li and Goad (2024) in their exposure group, despite similar materials, methods, and sample sizes. One might infer that the higher IPE detection rate in Li and Goad (2024) resulted from the exposure material being presented in participants' L1 (English), whereas English was an L2 for the current participants despite their advanced proficiency. However, this interpretation is not supported by responses to follow-up question 1: 75% ($n = 15$) of Thai-ex and 60% ($n = 12$) of Spanish-ex participants indicated that they had noticed (some of) the Mandarin speaker's plural errors in the exposure dialogue. This rate is nearly identical to the 75% ($n = 15$) of participants who noticed plural errors in Li and Goad (2024).

In addition, given the uniform advanced L2 English proficiency of all participants in the present study and the prediction of the CEM that prior L2 knowledge exerts either neutral or positive effects on additional language learning (Flynn et al., 2004), the morphological differences between English and Mandarin plurals did not appear to prevent participants from detecting the IPE. Thus, L1 background differences alone cannot fully explain the lower IPE detection rate compared with Li and Goad's (2024). A key distinction between the two studies lies in participants' language backgrounds: all participants in the present study knew only their L1 and English, whereas most participants in Li and Goad (2024) had some knowledge (below intermediate level) of one or more additional

European languages. Research indicates that richer multilingual experience can enhance cognitive abilities, metalinguistic awareness, and abstract language understanding (Bialystok, 2010; Cenoz, 2001; Jessner, 2008, 2012, 2014; Moore & Coste, 2006; Ransdell et al., 2006). It is therefore plausible that the more limited foreign language learning experience of the Thai-ex and Spanish-ex participants contributed to their lower IPE detection rate relative to participants in Li and Goad (2024).

8. Conclusion

This study investigated whether language learners, whose L1 is more similar to a foreign language, exhibit a superior ability to detect and utilize IPE presented in their L2 compared to those whose L1 is less similar to the target foreign language. Specifically, it examined L1-Thai and L1-Spanish participants with advanced L2 English proficiency. The study focused on the learning of 12 Mandarin nouns presented in singular form. Approximately half of the participants (Thai-ex and Spanish-ex groups) were exposed to an English dialogue providing IPE about the unmarked plural feature in Mandarin. Post-learning assessments evaluated participants' knowledge of the trained words and their use of IPE through singular and plural prompts over 2 consecutive days.

Findings indicated that although the majority of both Thai and Spanish participants noticed plural errors in the exposure material, only six Thai-ex and three Spanish-ex participants successfully linked these errors to the unmarked plural feature in Mandarin and incorporated the IPE in subsequent tests. The relatively higher IPE usage by Thai-ex participants compared with Spanish-ex participants may be attributed to their prior L2 English learning experiences. Thai and Mandarin both lack overt plural inflections, in contrast to the similarities between Spanish and English plural formation. Thai-ex participants' enhanced ability to recognize and apply IPE might be benefited from their history of encountering similar errors during L2 English learning, akin to the patterns produced by Mandarin speakers.

These findings suggest potential pedagogical advantages of incorporating accented input in language classrooms. Specifically, by exposing learners to authentic language productions containing IPE, teachers can foster a deeper understanding of grammatical structures and facilitate cross-linguistic knowledge transfer. Encouraging exposure to diverse linguistic varieties can enhance learners' linguistic flexibility and cultural competence, thereby enriching their overall language learning experience. Moreover, because learners whose L1 shares morphological similarities with the target language may benefit significantly from IPE, teachers might incorporate targeted IPE materials tailored to learners' specific L1 backgrounds. Such materials could enhance comprehension and retention of linguistic nuances.

Although the study sheds some light on the potential benefits of leveraging IPE in foreign language learning, it has several limitations. First, it focused only on the learning of Mandarin nouns in singular contexts, which may restrict the generalizability of the findings to other linguistic features or structures. Second, the relatively small sample size and the participants' specific language backgrounds may limit the applicability of the results to broader learner populations with diverse linguistic profiles and proficiency levels. Future research should explore the efficacy of IPE across a wider range of

linguistic features and contexts to provide a more comprehensive understanding of its role in language learning.

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Consent to Participate

The participants signed a consent form and volunteered to participate in the study.

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Declaration of Conflicting Interests

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Compliance with Ethical Standards

This study compliance with ethical standards. All the participants were adults.

Notes

1. Learners' ability to identify, analyze, and manipulate language forms.
2. All participants reported no formal or informal education in Chinese (Mandarin). Although studying in England provided opportunities to hear Chinese students, they had no Chinese friends.
3. A pilot study with six L1-Spanish and five L1-Thai adults who had advanced L2 English proficiency (IELTS 7.5 or 8.0) showed that all participants reported noticing some of the target IPE—errors using singulars in plural contexts—in the same English exposure material used in the main study.
4. A maxi test in this paper refers to a test that includes all of the trained words.
5. For these trials, the two pictures were combined as follows: (a) single non-target object (left) + single non-target object (middle); (b) single non-target object (left) + multiple non-target objects (middle); (c) multiple non-target objects (left) + single non-target object (middle); and (d) multiple non-target objects (left) + multiple non-target objects (middle).

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Appendix

Exposing Material

Contextually plural nouns produced without *s* inflection are underlined.

1. Liu Hua: Excuse me, I'm looking for the Student Service Centre. Do you know where it is?
2. Jerry: Yes. I'm heading there now. We can go together if you like.
3. Liu Hua: That would be great! Thank you!
4. Jerry: So, are you a new student at UCL?
5. Liu Hua: Yes, I am. I just arrived in London. What about you?
6. Jerry: I'm a 3rd year student. Where are you from?
7. Liu Hua: I'm from China. This is my first time abroad. What about you? Are you a British? I mean, you have blue eye, which is just like what I saw in western movie.
8. Jerry: Haha, you're right! I grew up in London, so I know every corner of this city. I can show you around if you like.
9. Liu Hua: Brilliant! Then I assume you know where I can buy some food, like real coffee bean and some fresh fruit. Oh, and some snack too. I love potato chip.
10. Jerry: You can find all that stuff in local supermarkets, like TESCO and Morisons. Where do you live?
11. Liu Hua: I live in Camden Town. Not far from campus. I haven't had time to explore nearby place yet. Are there many shop near where I live?
12. Jerry: Lucky you – your place is within walking distance of the campus. I know that TESCO and Waitrose are both close to where you live.
13. Liu Hua: What about kitchen item, like plate?
14. Jerry: I usually go to Wilko. Things are very cheap there. And you can find almost everything you need for your kitchen in that one store. Do you live alone or do you share an apartment with others?
15. Liu Hua: I live alone. Though I'd like to have some dog, but the apartment manager doesn't allow me to have any pet.

16. Jerry: Yeah, that's pretty common in London. It's for sanitary and safety reasons. Is your apartment a studio?
17. Liu Hua: What do you mean by a studio? I have four room: one bedroom, a living room, a kitchen, and a bathroom. Is that a studio?
18. Jerry: Yes, that's a studio. It must be spacious for you since you live alone.
19. Liu Hua: Definitely! And it is totally empty now. I mean, literally. So I'm thinking about buying some flower to brighten it up.
20. Jerry: Good idea! You can find flowers at the supermarket or in a flower shop.
21. Liu Hua: Great! How about cinema and bookstore or maybe a library? I love watching film and reading. They are definitely two of my favourite hobby.
22. Jerry: There are lots of cinemas and bookstores downtown. The ones I usually go to are on Gower Street, very close to the campus. And there's a library on campus, right next to the Student Service Centre.
23. Liu Hua: Sounds good! I'm planning on visiting some nearby city during the summer break. Do you have any suggestion where I should go?
24. Jerry: Well, you could begin with Cambridge if you are interested in visiting top universities.
25. Liu Hua: Is it far from London?
26. Jerry: Not at all. It takes about 1 hours to get there by train or bus. There are many interesting places to visit in London too.
27. Liu Hua: I heard that Buckingham Palace is famous.
28. Jerry: Yeah, it's where the royal family live. And there are some other interesting places in London too, like the London Zoo in Outer Cir. My family and I used to go there a lot when I was a kid.
29. Liu Hua: A zoo? I used to visit the zoo in my hometown with other kid from my school. It was fun. I especially love the elephant, but usually you can't find them in an urban zoo. It's amazing that their trunk are so flexible and they can pick up so many different object. How about the London Zoo? Are there any exotic animal there?
30. Jerry: Of course! It's a big zoo. There are elephants there too. And it is close from the city centre. Actually, my friends and I are planning on going there next weekend. You can join us if you like.
31. Liu Hua: That's wonderful! I'd love to, thank you!
32. Jerry: You're welcome. Okay, here we are. You see the building on the left-hand side? That's the Student Service Centre.
33. Liu Hua: Is it the one behind the wall?
34. Jerry: Yup. And that's the library beside it.
35. Liu Hua: Thank you so much! By the way, my name is Liu Hua.
36. Jerry: I'm Jerry. Nice to meet you Liu Hua.
37. Liu Hua: It's nice to meet you Jerry. This is my e-mail. I'll see you next weekend!
38. Jerry: Great, see you then!

Trained words (transcribed in IPA; a grave accent indicates tone 4)

[pèi] “cup” [bìn] “fan” [màu] “cat” [sùn] “ca”
[tòu] “plum” [dàn] “star” [màn] “hat” [lìn] “tree”
[kài] “car” [gòu] “book” [nèi] “bag” [lùn] “mo”